

论文

1. 《Logic Density Pruning: Mitigating Overthinking via Step-Level Chain of Thought Compression》

作者：杜恒铖，杨植景，陈俐宏，张晖；会议名称：NeurIPS 2026（ccf-a，Under Review）；时间：2026 年 5 月

Logic Density Pruning: Mitigating Overthinking via Step-Level Chain of Thought Compression

Hengcheng Du, Zhijing Yang, Lihong Chen, Hui Zhang

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Abstract:
Large reasoning models often improve on complex tasks by generating longer Chains-of-Thought (CoT). Yet longer CoTs are not necessarily more useful: redundant paraphrasing, loose explanations, and unproductive branches can substantially increase token usage and latency. This issue is often termed overthinking. Prior work attempts to control reasoning length via prompt engineering, teacher-driven rewriting and distillation, or training-based constraints, but it commonly lacks an interpretable metric over reasoning steps that attributes each reasoning step to the final answer, making it difficult to reliably balance compression and fidelity. To address this, we propose Logic Density Pruning (LDP). The LDP framework integrates two complementary signals—Mutual Information (MI) and attention citation—to quantify step importance via a unified criterion, logic density. We then perform intra-sample selection using Otsu adaptive thresholding. To reduce the risk of mistakenly removing truly critical steps, we introduce a step perturbation check that verifies whether removing a candidate step induces a significant shift in the model's subsequent generation distribution, and selectively recovers such steps. The resulting compressed CoTs are further used as supervision for fine-tuning, forming an end-to-end procedure from pruning to data construction and training. Notably, when our compressed CoTs are applied to Qwen3-8B, reasoning tokens are reduced by 80.2% (from 295.5 to 58.6) on CommonsenseQA, with only a 1.9% performance drop being observed. Ultimately, LDP provides an interpretable paradigm to compress complex reasoning without sacrificing essential task fidelity.

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Primary Area: Language and multimodal language models (e.g., text generation, summarization, VQA)
Secondary Area: Data-centric aspects of AI (e.g., data selection, active learning, curriculum learning, online learning)
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2. 《Estimate, Then Think: Prompt-Based Token Budgeting for Efficient Inference in Large Reasoning Models》

作者：杨植景，陈俐宏，肖国辉，杜恒铖，刘鹏，张乐梅，张晖；会议名称：AAAI 2027（ccf-a，在投）；时间：2026 年 7 月

Estimate, Then Think: Prompt-Based Token Budgeting for Efficient Inference in Large Reasoning Models

Zhijing Yang, Lihong Chen, Guohui Xiao, Hengcheng Du, Peng Liu, Lemei Zhang, Hui Zhang

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Abstract:
ETT predicts a problem-specific reasoning budget for prompt to mitigate Overthinking.

TLDR:
Existing efficient inference methods for Large Reasoning Models (LRMs) use fixed prompts, manual budgets, decoding controls, or model training. No existing method estimates a reasoning budget for each problem based on the ability of a given Target LRM. We address this gap with a Budget Estimator. It learns from successful Target LRM behavior and gives each problem a suitable budget. The predicted budget is a guidance value for the prompt, not an exact limit on response length. Based on this estimator, we propose Estimate-Then-Think (ETT). ETT has three stages: Training Data Construction, Budget Estimator Training, and Budget-Guided Inference. These stages build budget labels, train the estimator with GRPO, and apply its output through Zero-shot ETT (ETT-Z) or One-shot ETT (ETT-O). We test ETT on three LRMs and four reasoning benchmarks. ETT-O reduces token use by 33.02–46.70%. It also improves accuracy by 1.71–5.73 points and pass@3 by 3.27–4.11 points. On Qwen3-8B, ETT-O cuts token use by 46.70% and raises accuracy by 5.73 points. ETT therefore improves accuracy and efficiency with a separate budget guidance value for each problem.

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3. 《生育支持政策下青年婚育观研究》

作者：黄明秀，陈俐宏，汪正楠，钟一鸣；刊物名称：社会经济导刊；时间：2025 年 9 月

全文阅读

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摘要: 本研究聚焦成都市18-35岁青年群体的婚育状况, 探讨在网络媒体、经济压力等多重因素影响下, 青年婚育观念的变化及其现实困境。研究采用定量与定性结合的方法, 通过问卷调查和深度访谈, 分析了家庭环境、婚育教育、经济状况等因素对婚育意愿的影响。结果表明, 当代青年婚育观念呈现理性化、个性化与传统观念交织的特点, 婚育意愿整体积极但审慎。研究结论为政策制定和青年发展提供参考, 助力社会健康发展和人口结构改善。

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